

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction for effective data preparation (BL3, BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A SENARIO BASED QUESTIONS

<i>Criteria</i>	Understanding of Scenario (2)	Identification of Key Issues (2)	Application of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	<u>Total(10)</u>
<i>Weightage</i>	20%	20%	25%	20%	15%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

*I G SAI DEEPAK(192311432) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Deepak G—

① Monthly spend. Mean = $1001/8 = 125.133$
std dev = 166.53

Customer	Spend	Z-score
C ₁	20	-0.63
C ₂	35	-0.54
C ₃	38	-0.52
C ₄	500	+2.25
C ₅	25	-0.60
C ₆	45	-0.48
C ₇	38	-0.52
C ₈	300	+1.05

② Categorical Data Standardization :

① Standard Mapping:

Customer	Original	Standardized
C ₁	M	male
C ₂	Male	male
C ₃	F	female
C ₄	female	female
C ₅	male	male
C ₆	Female	female
C ₇	M	male
C ₈	Female	female

② Covariance

① Covariance

Mean

Mean (

Cov (

Customer

U₁

U₂

U₃

U₄

U₅

U₆

U₇

U₈

Cov

② Fre

Ag

Inco

Purcha

2 male

Cred

On

A

FM

② Covariance & Data Reduction in E-Commerce Analytics

Ⓐ Covariance Analysis:

$$\text{Mean} = 645 / 8 = 80.625$$

$$\text{Mean (purchase-Value)} = 1590 / 8 = 198.75$$

$$\text{Cov}(X, Y) = \sum (x_i - \bar{x})(y_i - \bar{y}) / n$$

Customer	Income(x)	$x - \bar{x}$	purchase(y)	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
U1	110	-40.625	100	-98.75	4011.25
U2	60	-20.625	150	-48.75	1005.47
U3	85	-15.625	210	11.25	-175.22
U4	120	39.375	320	121.25	4774.22
U5	45	-35.625	110	-88.75	3161.72
U6	75	-5.625	180	-18.75	105.47
U7	150	69.375	360	161.25	11185.55
U8	70	-10.625	160	-38.75	411.72

$$\text{Cov} = 24705.09 / 8 = 3088.14$$

Ⓑ Field Mean Std. dev

Age	39.375	10.22
Income	80.625	35.13
purchase-Value	198.75	88.52
Credit-Score	723.125	54.25
Orders	25.25	8.04
Avg_Rabnee	25.125	12.36
EMI	9500	4308.42

20/7/20

Co2 - ATL 2

G. Sai Deepak

1. Data Preprocessing in Retail Analysis

(A) Handling Missing Values:

(i) Missing Values Identified

* Age - Customer C₂

* Monthly Spend - Customer C₂

* Purchase - frequency - Customer C₅

(ii) & (iii) Imputation techniques applied:

Median Imputation:

<u>Missing field</u>	<u>Customer</u>	<u>Median Imputed</u>
Age	C ₂	36
Monthly Spend	C ₂	38
Purchase frequency	C ₃	10

(B) Outlier Detection & Treatment:

(i) Z-Score Calculation:

<u>Customer</u>	<u>Age</u>	<u>Z-Score</u>
C ₁	23	-0.80
C ₂	36	-0.27
C ₃	31	-0.47
C ₄	105	+2.57
C ₅	27	-0.64
C ₆	40	-0.10
C ₇	36	-0.27
C ₈	42	-0.2

Test Scores (range 40-100)
 Bin width = $(100 - 40) / 3 = 20$

Bin	Interval	Label	Student
1	40-59.99	Poor	S1, S2, S3
2	60-79.99	Average	S4, S5, S6
3	80-100	Excellent	S7-S10

(B) Equal frequency Binning (3-bins $\approx 4/3/3$)

Bin	Study Hours	Test Scores	Label
1	S1-S4	S1-S4	Low / poor
2	S5-S7	S5-S7	Medium / Average
3	S8-S10	S8-S10	High / Excellent

(C) Concept Hierarchy:

* Study Hours: Low (2-5 hrs); medium (6-8 hrs); High (9-11)

* Test Score: Poor (40-59) → Average (60-79) → Excellent (80-100)

Q:4: Normalization of: 200, 300, 400, 600, 1000

(a) Min-Max Normalization: $[\min=0, \max=1]$

$$x' = (x - \min) / (\max - \min), \min = 200, \max = 1000$$

X	200	300	400	600	1000
X'	0	0.125	0.25	0.5	1

⑥ Z-Score Normalization :

$$\text{Mean} = (200 + 300 + 400 + 500 + 1000) / 5 = 500$$

$$\text{Variance (population)} = \sum (x - \text{mean})^2 / n = 40000 / 5 = 8000$$

$$\text{Std dev} = \sqrt{8000} = 89.44$$

X	200	300	400	500	1000
Z	-1.661	-0.707	-0.354	0.354	1.768

Q5. Data Transformation in Online Shopping Behaviour Analysis :

① Equal-frequency Binning (3 bins, 4/3/3 split)

Bin	Customers	Values
1 (Short)	C1-C4	15, 30, 45, 60
2 (Medium)	C5-C7	120, 300, 600
3 (Long)	C8-C10	900, 1800, 3600

② Z-Score : $\text{Mean} = 700 / 10 = 70$

$$\text{Variance} = 11901.06 / 10 = 1190.106 \rightarrow \text{Std dev} = 34.5$$

Duration: 15 30 45 60 120 300 600 900 1800 3600

Z : -0.67 -0.66 -0.64 -0.63 -0.58 -0.41 -0.14 0.4 0.97 2.62

③ Min-Max Normalization :

$$X' = (x - 15) / (3600 - 15) = (x - 15) / 3585$$

Duration: 15 30 45 60 120 300 600 900 1800 3600

X' : 0 0.004 0.008 0.013 0.029 0.080 0.163 0.247 0.498 1.0

Cust	Age(2)	Income(3)	Purchase (2)	credit (2)	Active (2)	Balance (2)	2m (2)
U ₁	-1.51	-1.16	-1.12	-1.35	-1.28	-1.06	-1.01
U ₂	-0.53	-0.59	-0.55	-0.43	-0.65	-0.58	-0.52
U ₃	-0.45	0.13	0.12	-0.06	0.34	-0.01	-0.12
U ₄	1.24	1.12	1.29	1.42	1.21	1.20	1.28
U ₅	-1.02	-1.10	-1.00	-0.80	-0.90	-0.90	-0.93
U ₆	-0.04	-0.16	-0.21	-0.24	-0.16	-0.25	-0.13
U ₇	1.63	1.98	1.82	1.79	1.83	2.01	1.93
U ₈	-0.23	-0.30	-0.44	-0.33	-0.40	-0.42	-0.35

Question: 3 - Data Transformation & Discretization in Education Analytics.

(A) Equal-width Binning (3 bins)

(i) Study-Hours (range 2-11)

$$\text{Bin width} = (11 - 2) / 3 = 3$$

Bin	Interval	Label	Students
1	2-4.99	Low	S ₁ , S ₂ , S ₃
2	5-7.99	Medium	S ₄ , S ₅ , S ₆
3	8-11	High	S ₇ , S ₈ , S ₉ , S ₁₀